Ilankoon Pathiranage Pabasara Vikum Ilankoon

pabasara-ilankoon | pabasara-ilankoon | Portfolio | pvillankoon@gmail.com | +94 76 251 8343

Professional Summary

Electronic and Telecommunication Engineering undergraduate at General Sir John Kotelawala Defence University with a strong interest in **Embedded Systems, Artificial Intelligence, Machine Learning, Computer Vision, IoT and Wireless Communication**. Experienced in developing innovative engineering solutions through projects involving **AI-powered detection systems, embedded hardware integration, cloud-connected applications and long-range communication networks**. Proficient in Python, C++, Raspberry Pi, ESP32, OpenCV, YOLO and related technologies, with strong analytical, problem-solving and teamwork skills demonstrated through multidisciplinary engineering projects.

Education

General Sir John Kotelawala Defence University — **Jan. 2024 – Present**
BSc. (Hons) in Electronic and Telecommunication Engineering

University of Colombo School of Computing — **Sep. 2024 – Present**
BSc. in Information Technology (External)

St. Anne's College, Kurunegala — **Jan. 2013 – Feb. 2022**
GCE Advanced Level (Physical Science Stream)
GCE Ordinary Level

Relevant Coursework

- Signals and Systems
- Electronic Systems
- Communication Theory
- Control Systems
- Electromagnetics
- Electronic Circuits
- Communication Networks
- Digital System Design
- Embedded Systems
- Digital Signal Processing
- Power Electronics
- Image Processing and Vision

Technical Skills

Programming	Python, C, C++, Arduino, Dart, JavaScript
AI/ML & Vision	Machine Learning, Deep Learning, OpenCV, YOLO, CNN, TensorFlow, Keras, MobileNetV2, Transfer Learning, Image Processing, Object Detection, Computer Vision, CVAT, Data Annotation
Embedded Systems	ESP32, STM32, Raspberry Pi, Arduino, Sensor Integration, Motor Control, Motor Drivers, PCB Designing, PCB Prototyping, Circuit Debugging
Robotics & Autonomy	Maze Solving Robots, Autonomous Navigation, Path Planning, ROS2, Sensor Fusion, Obstacle Detection, PID Control, Real-Time Embedded Systems, PCB Designing, Robot Control Systems
Cloud & Software	Firebase, FastAPI, REST APIs, Flutter, React.js, Tailwind CSS, Git, GitHub, Mobile App Development, Web Application Development
Communication	LoRa, GSM, Wi-Fi, Bluetooth, UART, SPI, I2C, IoT Systems
Design Tools	EasyEDA, Proteus, Solidworks, LTSpice, MATLAB, VS Code, Arduino IDE, Microchip Studio, Tindercad

Projects

Elevision | AI-Based Elephant Detection and Early Warning System

- Designed and developed a real-time AI-based railway safety system to detect elephants on or near railway tracks using computer vision and deep learning.
- Implemented YOLO-based object detection with OpenCV for real-time video frame analysis and wildlife detection.
- Built a web application for live monitoring, alert visualization and system status tracking.
- Developed a Flutter mobile application for real-time alerts and remote monitoring.
- Integrated Raspberry Pi edge computing, GSM and Firebase cloud services for real-time alerting and storage.

Technologies: Python, OpenCV, YOLO, CNN, Raspberry Pi, Firebase, GSM, IoT, Flutter, React.js

LankaMesh | LoRa-Based Disaster Communication System

- Designed and developed a decentralized emergency communication system for disaster environments without internet or cellular networks.
- Implemented LoRa-based long-range peer-to-peer communication for SOS alerts, messages and GPS data sharing.
- Designed embedded system architecture using ESP32-S3, LoRa RA-02 (SX1278) and GPS modules.
- Built a Flutter mobile application for emergency alert categorization and communication handling.
- Demonstrated scalable IoT-based disaster communication framework for resilient connectivity.

Technologies: ESP32-S3, LoRa RA-02, GPS, Flutter, IoT, Embedded Systems, Wireless Communication

FinFlow | Secure Personal Finance Manager

- Developed a cross-platform personal finance management application with secure user authentication and voice-assisted expense tracking.
- Designed and integrated secure RESTful APIs with OAuth 2.0 authentication and token-based authorization for protected client-server communication.
- Built a scalable backend using Node.js and Express.js to manage accounts, transactions, budgets and expense categories.
- Developed a Flutter mobile application featuring interactive dashboards, financial analytics, budget tracking and real-time expense management.
- Implemented secure client-server communication with authenticated API requests and efficient data synchronization.

Technologies: Flutter, Dart, Node.js, Express.js, JavaScript, REST APIs, OAuth 2.0, JWT, Speech-to-Text, Text-to-Speech

AgroVision AI | Intelligent Tomato Leaf Disease Detection System

- Developed a full-stack AI system for tomato leaf disease detection using deep learning and computer vision.
- Built a MobileNetV2 transfer learning model for multi-class plant disease classification.
- Developed web (React.js) and mobile (Flutter) applications for image upload and prediction results.
- Implemented FastAPI backend for authentication, prediction and database management.
- Integrated Firebase cloud storage for image dataset handling and prediction history tracking.

Technologies: Python, TensorFlow, Keras, OpenCV, FastAPI, React.js, Flutter, Firebase, CNN, MobileNetV2

Digital Communication Simulator

- Developed a web-based simulation platform to model and analyze digital communication systems.
- Implemented modulation schemes including ASK, QPSK, 16-QAM and OFDM for system analysis.
- Simulated AWGN channel effects to evaluate signal degradation and noise impact.
- Built BER vs SNR performance evaluation tools for communication system comparison.
- Designed interactive signal visualization tools using Plotly and Matplotlib.

Technologies: Python, Flask, NumPy, SciPy, Plotly, Matplotlib, HTML, CSS, JavaScript, DSP

Mars Robot Competition | Autonomous Robotics System

- Designed and developed an autonomous robotic system for competition-based navigation and object handling tasks.
- Implemented sensor fusion using IR sensors, ultrasonic sensors, gyroscope and color sensors.
- Developed embedded control system using ESP32-S3 for real-time decision-making and motor control.
- Designed mechanical system using SolidWorks and 3D printing for structural fabrication.
- Implemented encoder-based motor control using TB6612FNG driver for precise movement control.

Technologies: ESP32-S3, C/C++, IR, Ultrasonic, MPU6050, TCS34725, SolidWorks, Embedded Systems

ROSCO'25 Competition Robot | Autonomous Line and Wall Following Robot

- Built an autonomous robot capable of line following, wall following and ramp navigation tasks.
- Implemented sensor fusion using VL53L0X ToF sensors, IR array and MPU6050 IMU.
- Developed embedded control system using ESP32-S3 and TB6612FNG motor driver.
- Designed chassis using laser-cut acrylic and custom 3D-printed mechanical parts.
- Integrated encoder-based feedback system for accurate motion control and stability.

Technologies: ESP32-S3, VL53L0X, IR Sensors, MPU6050, TB6612FNG, SolidWorks, Embedded Systems

FeelFill | Smart Blind-Friendly Measuring Cup (Ongoing)

- Developing a portable assistive device for blind users to measure liquid levels using non-contact capacitive sensing.
- Implementing external capacitive detection through cup walls for real-time liquid level monitoring without liquid contact.
- Using ESP32-C3 Mini for sensor processing, calibration, and multi-level detection.
- Designing vibration and buzzer feedback system with different alerts for fill levels.
- Building a compact clip-on device with rechargeable battery and low-power embedded design.

Technologies: ESP32-C3 Mini, Capacitive Sensing, Embedded Systems, PCB Design, IoT, Vibration Motor, Buzzer

Achievements

Champions	Target Sprint Shooting Competition - 2025
Precision Shooter Award	MSSC Shooting Championship - 2025
KDU Colours	Air Rifle - 2025

Certifications

AI/ML Engineer	2026, SLIIT; Machine learning, supervised and unsupervised learning, deep learning, neural networks, TensorFlow, Keras and model evaluation techniques.
Machine Learning Concepts	2026, Microsoft; Machine learning fundamentals, supervised and unsupervised learning, model evaluation and AI concepts.
PCB Designing Workshop	2025, IET KDU & Sri Lanka Institute of Robotics (SLIR); PCB schematic design, component placement, PCB layout, design rule checking (DRC) and Gerber file generation.
MATLAB Fundamentals	2024, MATLAB Training Program; Numerical computing, matrix operations, data visualization, scripting and engineering simulation techniques.

Leadership & Volunteering

Chairman: IEEE ComSoc KDU	June 2026 – Present
Finance Team Lead: GENESIZ'26	May 2026 – Present
Finance Committee Member: ERIC KDU	Feb. 2026 – Present
Program Team Member: IEEE ComSoc KDU	June 2024 – Present
Design Committee Member: Mathematical Society KDU	Feb. 2025 – Feb. 2026
PR Team Member: ERIC KDU	Feb. 2025 – Feb. 2026

Professional Affiliations

IET Student Member: Institution of Engineering and Technology (UK)

IESL Student Member: Institution of Engineers Sri Lanka

References

Prof. T.L. Weerawardane
Professor
Dept. of Electronic and Telecom. Engineering
General Sir John Kotelawala Defence University
Email: tlw@kdu.ac.lk
Tele: +94 77 444 1419

Prof. S.U. Dampage
Professor
Dept. of Electronic and Telecom. Engineering
General Sir John Kotelawala Defence University
Email: dampage@kdu.ac.lk
Tele: +94 70 340 5405